Wen Sheng
WS2840

#1. There is arbitrage opportunity.

no-arbitrage forward rate $F^* = S_0 \cdot \frac{1+r_{USD}}{1+r_{EUR}} = 1.149 \cdot \frac{1+4.46\%}{1+3.35\%} \approx 1.1613$ USD/EUR

$F^* = 1.16134 < 1.16485$

Therefore EUR foward rate is overpriced.

To arbitrage:

1. Borrow $1USD at 4.46%
2. convert to euro $\frac{1}{1.149} = 0.87032$ EUR
3. Invest EUR for 1 year at 3.35%.

$0.87032 \times (1+3.35\%) = 0.89948$ EUR

4. Sell 0.89948 EUR 1-year forward at 1.16485 USD/EUR at T0

5. In one year, receive from forward:

$0.89948 \times 1.16485 = 1.04776$ USD

6. Repay USD loan $1 \times (1+4.46\%) = 1.0446$ USD

7. You get risk-free profit:

$1.04776 - 1.0446 = 0.00316$ USD (0.316% return on $1 investment)

#2. $F = S_0 \cdot e^{(r_{USD} - r_{EUR}) \cdot T}$

$= 1.148 \cdot e^{(4.023\% - 2.62\%) \cdot \frac{3}{12}}$

≈ 1.1520 USD/EUR

The 3-month forward exchange rate is 1.1520 USD/EUR

#3.

(i) A negative interest rate means lender pays the borrower.

(ii) For negative deposit rate, bank's reserve balance at the central bank shrink over time

(iii) This encourages banks to reduce excess reserves and push overnight and other short-term rate lower.

(iv) Lower rates will stimulate lending and investment, and weaken the currency.

#4.

$$F = S_0 e^{(r-d)T}$$

① 4.5% interest

$$F = 100 \cdot e^{4.5\% \cdot \frac{6}{12}} \approx 102.28$$

② 3% dividend yield

$$F = 100 \cdot e^{(4.5\% - 3\%) \cdot \frac{6}{12}} \approx 100.75$$

③ 4% dividend yield

$$F = 100 \cdot e^{(4.5\% - 4\%) \cdot \frac{6}{12}} \approx 100.25$$

④ 5% dividend yield

$$F = 100 \cdot e^{(4.5\% - 5\%) \cdot \frac{6}{12}} \approx 99.75$$

#5.

(i) Forwards are private contracts between two parties;
Futures are exchange traded.

(ii) Forwards are non-standard contract; Futures are standard contract.

(iii) Forwards usually have 1 specified delivery date;
Futures have a range of delivery dates

(iv) Forwards settled at end of contracts; Futures settled daily

(v). For Forwards, delivery of final cash settlement usually occurs;
Future contracts usually close out prior to maturity.

(vi) Forwards have some credit risk; Futures virtually have no credit risks.

#6. Margin call dates: 09/11/2024, 09/12/2024, 09/17/2024, 9/24/2024, 10/03/2024
10/07/2024, 03/09/2026, 03/11/2026, 03/12/2026, 03/13/2026
03/18/2026, 04/28/2026, 04/29/2026, 05/04/2026, 05/18/2026,
09/08/2026, 09/09/2026, 09/10/2026, 09/15/2026

ate	Close	Daily Gain ($)	Cumulative Gain ($)	Margin Balance ($)	Margin Call ($)
09/11/24	63.94	-1680	-1680	14572	1680
09/12/24	64.83	-1780	-3460	14472	1780
09/17/24	65.64	-1080	-5080	14632	1620
09/24/24	66.69	-1560	-7180	14152	2100
10/03/24	68.47	-3760	-10740	12692	3560
10/07/24	69.61	-2980	-13020	13972	2280
03/09/26	71.17	-3860	-16140	13132	3120
03/11/26	72.31	-5880	-18420	13972	2280
03/12/26	75.17	-5720	-24140	10532	5720
03/13/26	76.32	-2300	-26440	13952	2300
03/18/26	77.86	-3000	-29520	13172	3080
04/28/26	78.68	-840	-31160	14612	1640
04/29/26	79.49	-1620	-32780	14632	1620
05/04/26	83.72	-8380	-41240	7792	8460
05/18/26	85.32	-3800	-44440	13052	3200
09/08/26	87.14	-3360	-48080	12612	3640
09/09/26	89.32	-4360	-52440	11892	4360
09/10/26	93.93	-9220	-61660	7032	9220
09/15/26	95.51	-5820	-64820	13092	3160

#7

Mean : 0.10220/0

Variance : 0.000289/5

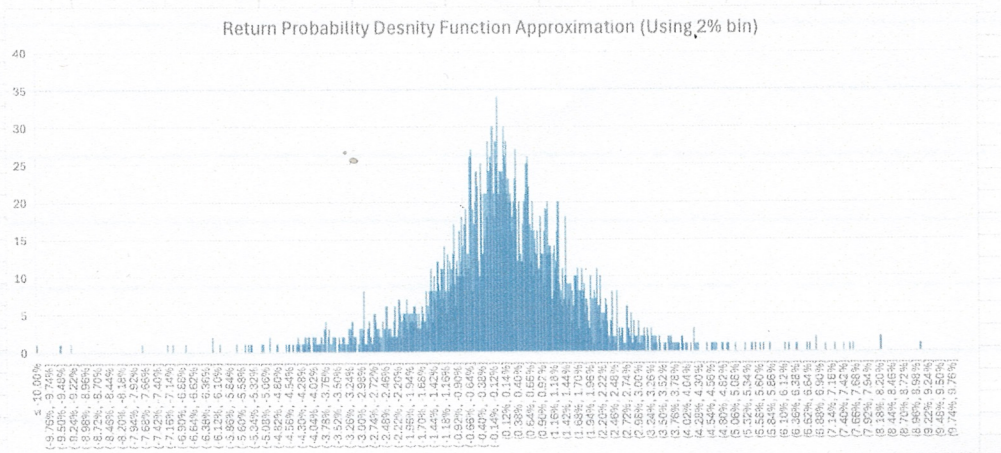
STD : 1.7015%

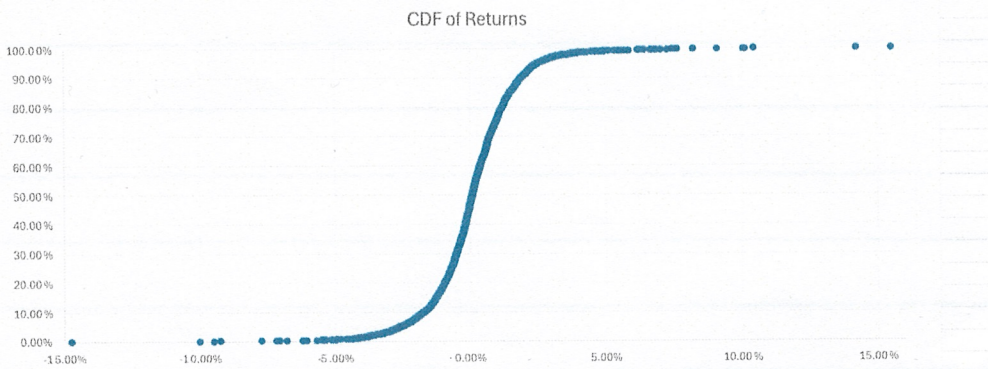
Excess kurtosis : 9.20

Kurtosis : 12.20

Skewness : 0.31613

Mean	0.1022%
Variance	0.000289508
Standard Deviation	1.7015%
Mean Absolute Deviation	1.1611%
Excess Kurtosis	9.20
Kurtosis	12.20
Skewness	0.316133258
Min Return	-14.74%
Max Return	15.51%

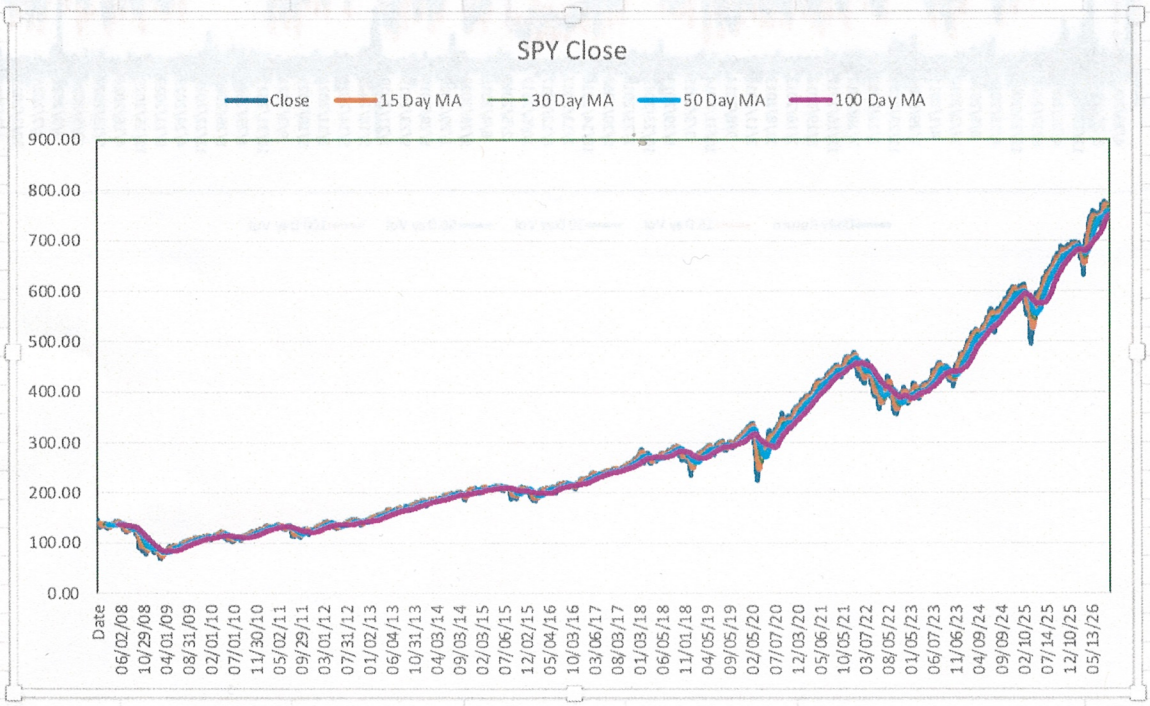




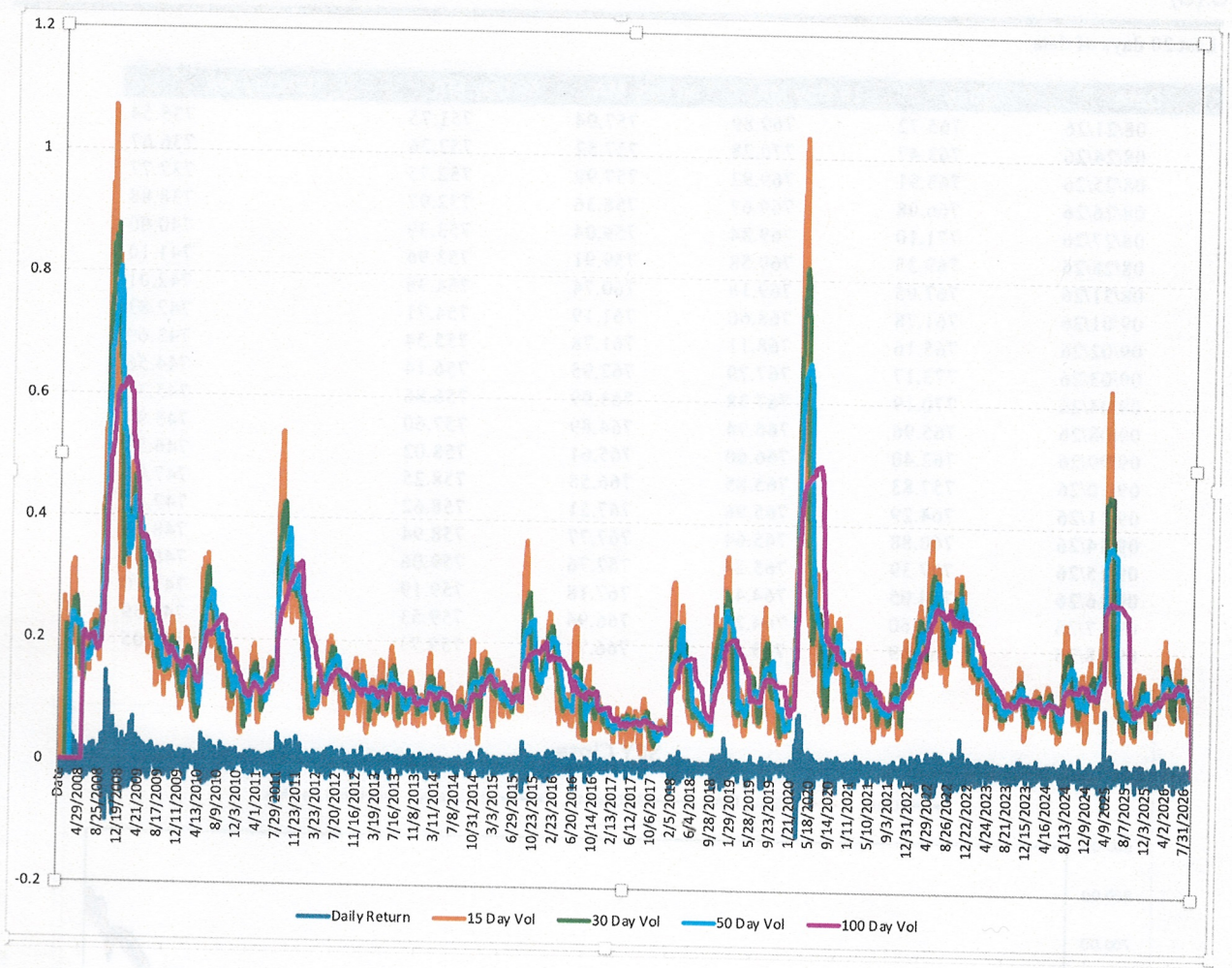
8.(a)

Last 20 days of data

Date	Close	15 Day MA	30 Day MA	50 Day MA	100 Day MA
08/21/26	765.72	769.89	757.04	751.75	735.54
08/24/26	763.47	770.28	757.52	752.26	736.67
08/25/26	765.91	769.92	757.99	752.75	737.77
08/26/26	766.08	769.67	758.36	752.97	738.88
08/27/26	771.10	769.84	759.04	753.39	740.00
08/28/26	769.35	769.58	759.91	753.96	741.10
08/31/26	767.05	769.18	760.74	754.36	742.01
09/01/26	761.78	768.60	761.19	754.71	742.83
09/02/26	765.16	768.11	761.78	755.34	743.69
09/03/26	773.17	767.79	762.95	756.14	744.56
09/04/26	770.19	767.38	763.99	756.86	745.31
09/08/26	765.96	766.94	764.89	757.60	745.97
09/09/26	762.40	766.60	765.61	758.02	746.58
09/10/26	757.83	765.85	766.55	758.25	747.06
09/11/26	764.29	765.96	767.31	758.62	747.61
09/14/26	760.88	765.64	767.77	758.94	748.18
09/15/26	757.39	765.24	767.76	759.06	748.64
09/16/26	754.05	764.45	767.18	759.19	749.10
09/17/26	762.60	764.21	766.94	759.53	749.59
09/18/26	761.69	763.59	766.71	759.73	750.05



8.(b)


1.2
1
0.8
0.6
0.4
0.2
0
-0.2
Daily Return
15 Day Vol
30 Day Vol
50 Day Vol
100 Day Vol

#9

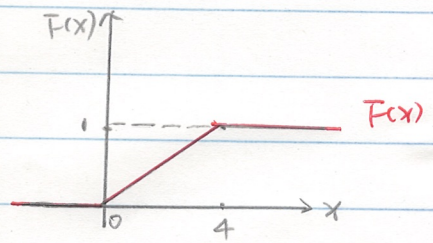
$X \sim \text{Uniform}(0, 4)$

$E[X] = \frac{4+0}{2} = 2$

$\text{Var}[X] = \frac{1}{12}(4-0)^2 = \frac{16}{12} = \frac{4}{3} \approx 1.33$

$\text{Stdev}[X] = \sqrt{\text{Var}[X]} = \sqrt{\frac{4}{3}} \approx 1.15$

CDF: $F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{4}, & 0 \le x \le 4 \\ 1, & x > 4 \end{cases}$



#10.

$X \sim N(1, 2^2)$

$Y \sim N(3, 1^2)$

$\rho_{xy} = 0.4$

(1) $E[X+Y] = 1+3 = 4$

$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2\,\text{Cov}(X, Y)$

$= 2^2 + 1^2 + 2 \cdot 0.4 \cdot 2 \cdot 1$

$= 4 + 1 + 1.6 = 6.6$

$\text{Stdev}(X+Y) = \sqrt{\text{Var}(X+Y)} = \sqrt{6.6} \approx 2.57$

(2) The distribution of $X+Y$ is unknown given the information

(3) If X and Y are jointly normal, then $X+Y \sim N(4, 6.6)$

(4) If X and Y are not jointly normal then it can't imply $X+Y$ is normal. And the distribution is unknown. But $E[X+Y] = 4$, $\text{Var}[X+Y] = 6.6$ still holds.

#11.

A_i : admit to school i, $P_r(A_i) = 7\%$

Assume $A_1, A_2 \dots A_n$ are independent

B : Admit into At leaset one school

$$P_r(B) = 1 - \prod_{i=1}^{n} (1 - P_r(A_i))$$

$$= 1 - (1 - 7\%)^n \geq 98\%$$

So we have

$$(93\%)^n \leq 2\%$$

$$n \geq 53.91$$

Therefore,

We need to apply at least 54 schools

Pledge

I pledge that this represents my own work and did not copy solutions and did not use AI.